1202SCG Calculus 1
1020ENG Engineering Mathematics 2

Week 3 Lecture

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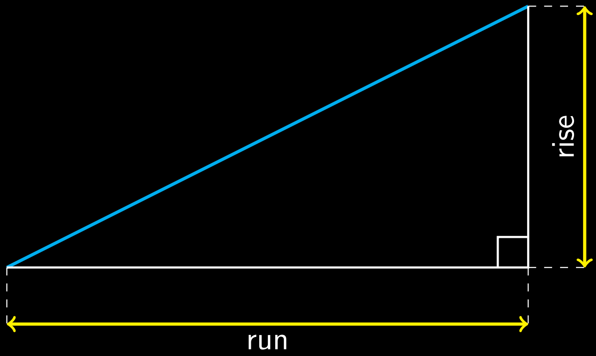
Rates Are Everywhere

Many everyday quantities describe how one quantity changes relative to another.

- **Speed:** kilometres/hour
- **Pay:** \$/year or \$/hour
- **Medication:** tablets/day
- **Heart rate:** beats/minute
- **Water flow:** litres/minute
- **Data transfer:** megabytes/second (megabits/second)
- **COVID infection growth:** new infections/day

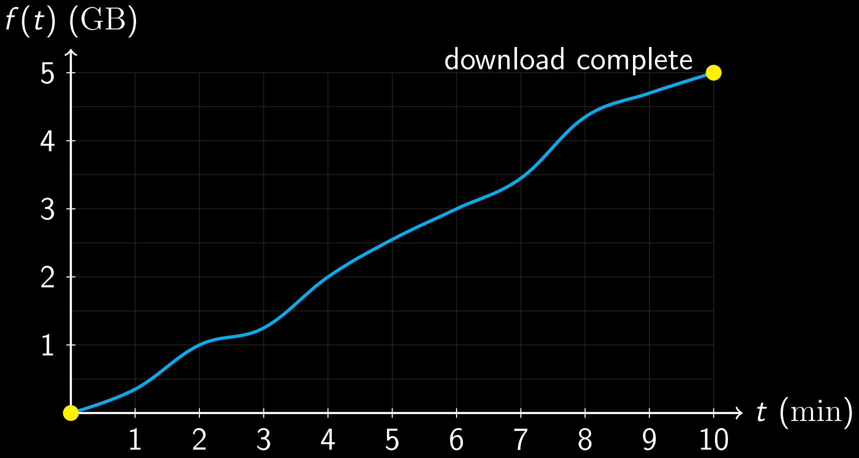
Revision of Gradient



$$\text{gradient} = \frac{\text{rise}}{\text{run}}$$

Downloading a Large File

Let $f(t)$ be the amount of a 5 GB file downloaded after t minutes.



Average Rates of Change

An average rate of change is measured between two points.

$$\text{Average rate of change} = \frac{f(b) - f(a)}{b - a},$$

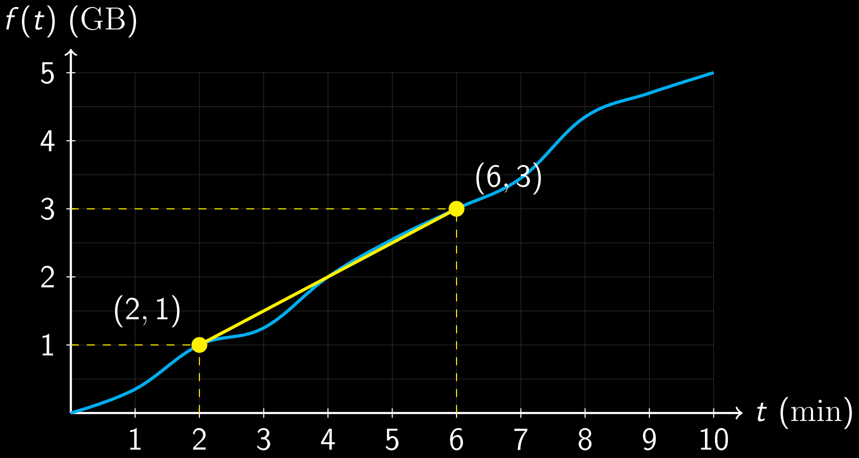
$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

$$\text{Average download speed} = \frac{f(t_2) - f(t_1)}{t_2 - t_1}.$$

The average rate of change is the slope of the line joining the two points.

Average Download Speed

Calculate the average download speed between $t = 2$ and $t = 6$ minutes.



Average Download Speed: Solution

Between $t = 2$ and $t = 6$,

$$f(2) = 1 \text{ GB}, \qquad f(6) = 3 \text{ GB}.$$

$$\text{Average download speed} = \frac{f(6) - f(2)}{6 - 2}$$

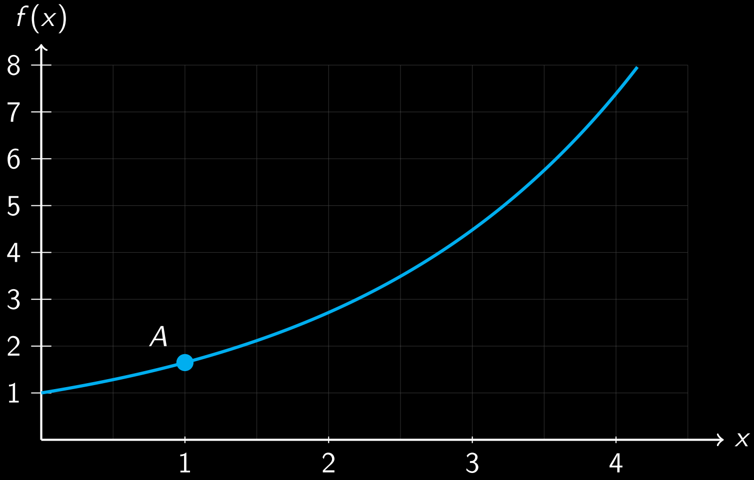
$$= \frac{3 - 1}{4}$$

$$= 0.5 \text{ GB/min}.$$

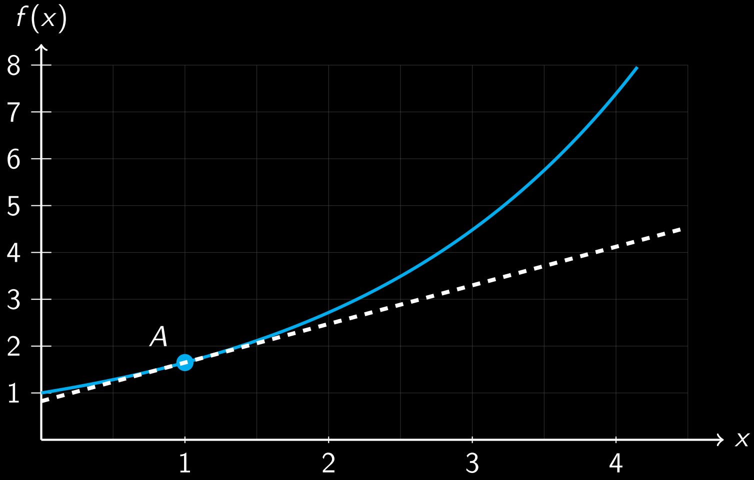
$$0.5 \text{ GB/min} = 500 \text{ MB/min} = 8.33 \text{ MB/s} = 66.7 \text{ Mb/s}.$$

Uppercase B denotes bytes; lowercase b denotes bits.

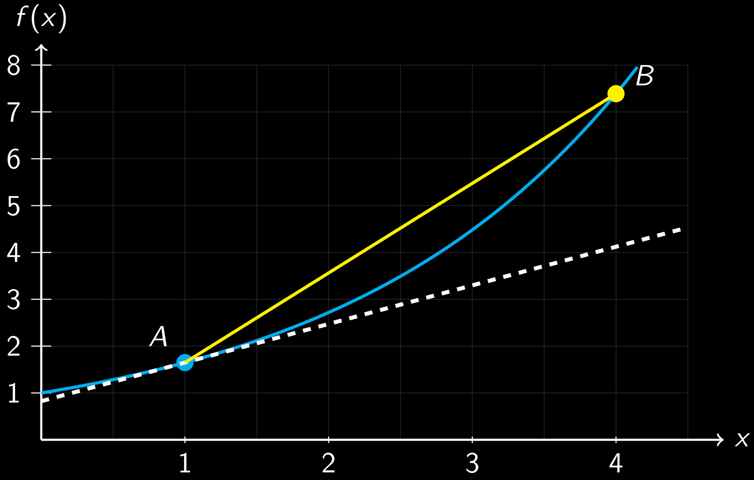
Rates of Change



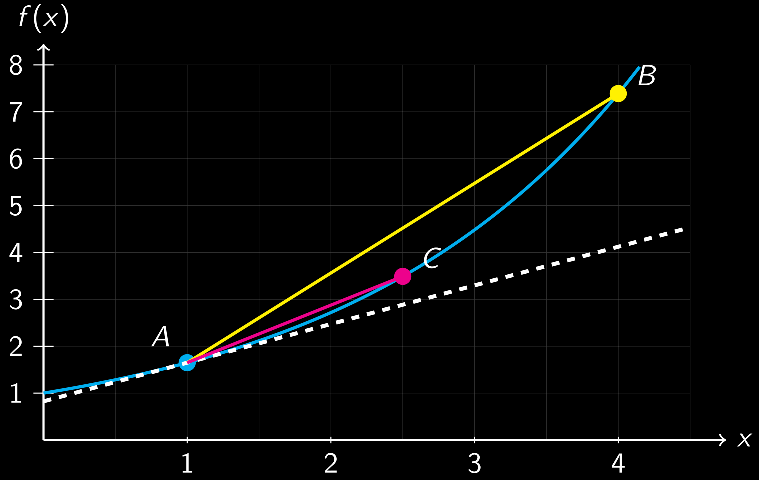
Rates of Change



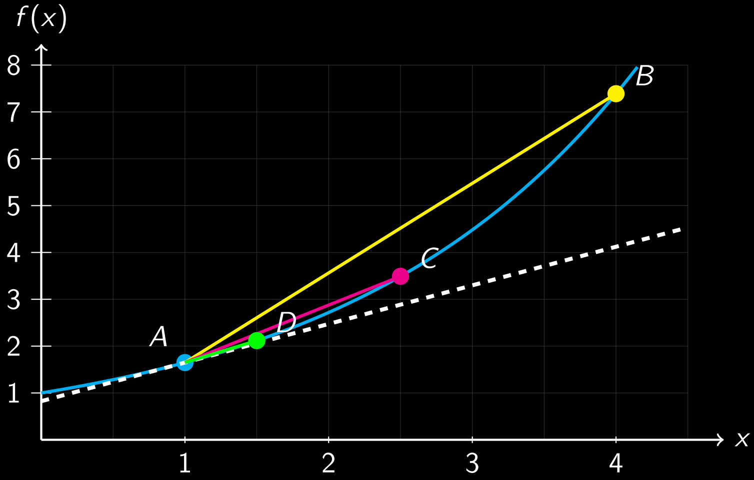
Rates of Change



Rates of Change

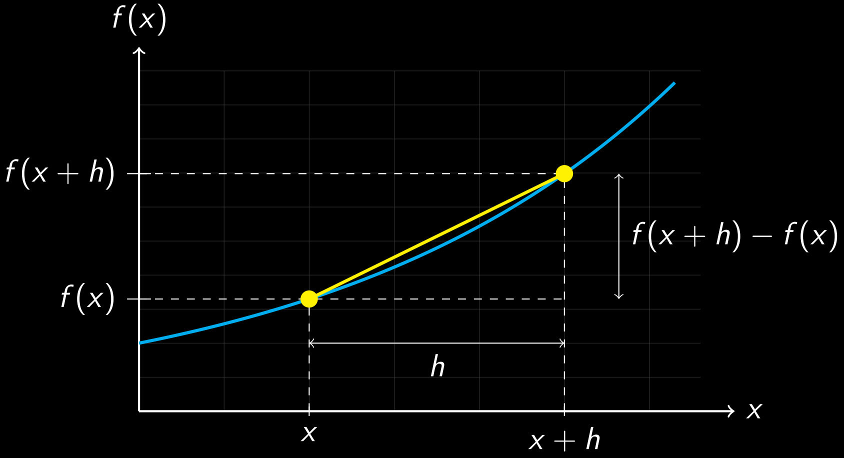


Rates of Change



Formal Definition of a Derivative

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$



Derivative Notation

The first derivative may be written in several equivalent ways:

$$y', \qquad f'(x), \qquad \frac{dy}{dx}, \qquad \frac{d}{dx}\big[f(x)\big].$$

Higher-Order Derivatives

$$\text{Second derivative:} \qquad y'', \quad f''(x), \quad \frac{d^2y}{dx^2},$$

$$\text{Third derivative:} \qquad y''', \quad f'''(x), \quad \frac{d^3y}{dx^3}.$$

Words That Mean Derivative

In this context, the following descriptions refer to the same idea:

- derivative;
- slope;
- gradient;
- instantaneous rate of change.

Each describes how quickly a quantity is changing at a particular point.

First Principles: $y = f(x) = x^2$

$$
\begin{aligned}
\frac{dy}{dx} &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{h(2x+h)}{h} \\
&= \lim_{h \to 0} (2x+h) \\
&= 2x.
\end{aligned}
$$

First Principles: $y = f(x) = 1/x$

$$
\begin{aligned}
\frac{dy}{dx} &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\
&= \lim_{h \to 0} \frac{\frac{x-(x+h)}{x(x+h)}}{h} \\
&= \lim_{h \to 0} \frac{-h}{hx(x+h)} \\
&= \lim_{h \to 0} \left(-\frac{1}{x(x+h)} \right) \\
&= -\frac{1}{x^2}.
\end{aligned}
$$

Table of Derivatives

In the table below, n and c are constants.

Function y	Derivative $\frac{dy}{dx}$
c	0
x^n	nx^{n-1}
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
e^x	e^x
$\ln(x)$	$\frac{1}{x}$

Using the Power Rule

Example 1

$$y = \frac{1}{x^3}$$

$$= x^{-3}$$

$$\frac{dy}{dx} = -3x^{-4}$$

$$= -\frac{3}{x^4}.$$

Example 2

$$y = \sqrt{x}$$

$$= x^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$$

$$= \frac{1}{2x^{\frac{1}{2}}}$$

$$= \frac{1}{2\sqrt{x}}.$$

Expanded Table of Derivatives

In the table, $a > 0$ and $a \neq 1$.

Function y	**Derivative** $\frac{dy}{dx}$
$\tan(x)$	$\frac{1}{\cos^2(x)} = \sec^2(x)$
a^x	$a^x \ln(a)$
$\log_a(x)$	$\frac{1}{x \ln(a)}$
$\sinh(x)$	$\cosh(x)$
$\cosh(x)$	$\sinh(x)$
$\tanh(x)$	$\frac{1}{\cosh^2(x)} = \operatorname{sech}^2(x)$

Rules for Combining Derivatives

Let c be a constant and suppose that $f(x)$ and $g(x)$ are differentiable.

Sum rule: $$\frac{d}{dx}\big[f(x)+g(x)\big]=\frac{d}{dx}f(x)+\frac{d}{dx}g(x),$$

Difference rule: $$\frac{d}{dx}\big[f(x)-g(x)\big]=\frac{d}{dx}f(x)-\frac{d}{dx}g(x),$$

Constant multiple rule: $$\frac{d}{dx}\big[cf(x)\big]=c\frac{d}{dx}f(x).$$

Derivative Practice

Find $\dfrac{dy}{dx}$ for each function.

1. $y = 5x^7 - 3x^4 + 2x - 8$
2. $y = \dfrac{3}{x^4} - 2\sqrt{x}$
3. $y = 6\sin(x) - 4\cos(x)$
4. $y = 2\tan(x) + 5x^3$
5. $y = 4e^x - 3(2^x)$
6. $y = 5\ln(x) + 2\log_3(x)$
7. $y = 3\sinh(x) - 2\cosh(x)$
8. $y = 4\tanh(x) - 5x^2$

Derivative Practice: Answers

1. $\dfrac{dy}{dx} = 35x^6 - 12x^3 + 2$
2. $\dfrac{dy}{dx} = -\dfrac{12}{x^5} - \dfrac{1}{\sqrt{x}}$
3. $\dfrac{dy}{dx} = 6\cos(x) + 4\sin(x)$
4. $\dfrac{dy}{dx} = 2\sec^2(x) + 15x^2$
5. $\dfrac{dy}{dx} = 4e^x - 3(2^x)\ln(2)$
6. $\dfrac{dy}{dx} = \dfrac{5}{x} + \dfrac{2}{x\ln(3)}$
7. $\dfrac{dy}{dx} = 3\cosh(x) - 2\sinh(x)$
8. $\dfrac{dy}{dx} = 4\,\mathrm{sech}^2(x) - 10x$

Position, Velocity, and Acceleration

A delivery robot moves along a straight corridor. Its position is

$$s(t) = t^3 - 3t^2 + 2t + 5,$$

where s is measured in metres and t in seconds.

Determine:

1. the position of the robot at $t = 2$ and $t = 3$;
2. its average velocity between $t = 2$ and $t = 3$;
3. its instantaneous velocity at $t = 2$;
4. its instantaneous acceleration at $t = 2$.

Position and Average Velocity

$$
\begin{aligned}
s(2) &= (2)^3 - 3 \times (2^2) + 2(2) + 5 \\
&= 5 \text{ m} \\
s(3) &= (3)^3 - 3 \times (3^2) + 2(3) + 5 \\
&= 11 \text{ m} \\
\text{Average velocity} &= \frac{s(3) - s(2)}{3 - 2} \\
&= \frac{11 - 5}{1} \\
&= 6 \text{ m/s.}
\end{aligned}
$$

Instantaneous Velocity

Velocity is the derivative of position:

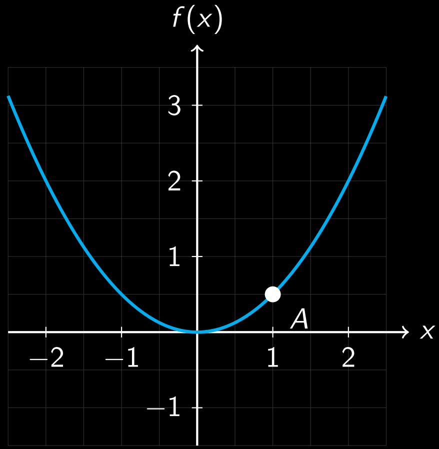
$$\begin{aligned} v(t) &= \frac{d}{dt}\left[t^3 - 3t^2 + 2t + 5\right] \\ &= 3t^2 - 6t + 2 \\ v(2) &= 3 \times (2^2) - 6 \times (2) + 2 \\ &= 2 \text{ m/s.} \end{aligned}$$

Acceleration

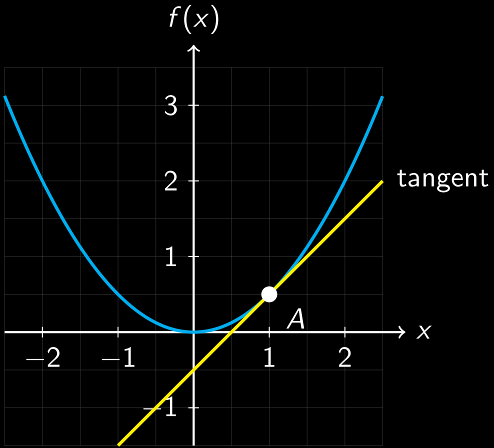
Acceleration is the derivative of velocity:

$$
\begin{aligned}
a(t) &= \frac{d}{dt}\left[3t^2 - 6t + 2\right] \\
&= 6t - 6, \\
a(2) &= 6 \times (2) - 6 \\
&= 6\ \mathrm{m/s^2}.
\end{aligned}
$$

Tangent and Normal Lines

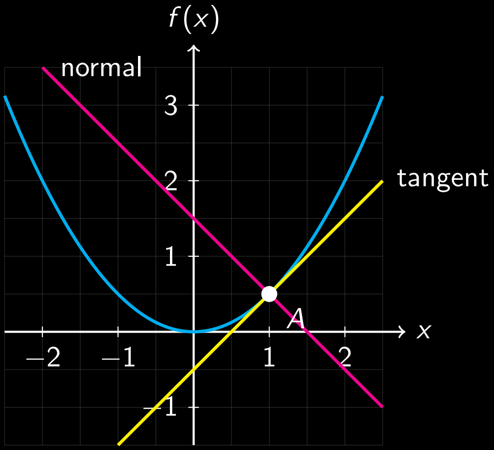


Tangent and Normal Lines



The tangent passes through A and has the same gradient as the function at A.

Tangent and Normal Lines



The normal passes through A and is perpendicular to the tangent.

Finding the Tangent Line

Find the tangent to $y = x^3$ at $x = 1$.

$$y = 1^3 = 1, \qquad A = (1, 1),$$

$$\frac{dy}{dx} = 3x^2,$$

$$m_{\text{tangent}} = \left.\frac{dy}{dx}\right|_{x=1} = 3.$$

Using $y - y_1 = m(x - x_1)$,

$$y - 1 = 3(x - 1)$$

$$y = 3x - 2.$$

Finding the Normal Line

Find the normal to $y = x^3$ at $x = 1$.
The normal is perpendicular to the tangent, so

$$m_{\text{normal}} = -\frac{1}{m_{\text{tangent}}}$$

$$= -\frac{1}{3}.$$

Using $A = (1, 1)$ and the point–slope form,

$$y - 1 = -\frac{1}{3}(x - 1)$$

$$y = -\frac{1}{3}x + \frac{4}{3}.$$

x-intercept of the Tangent Line

The tangent is $y = 3x - 2$.

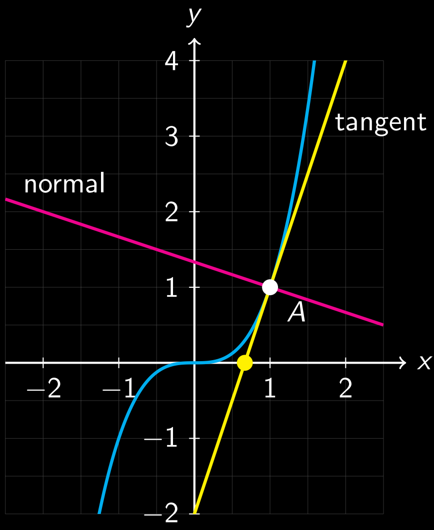
At the x-intercept, $y = 0$.

$$0 = 3x - 2$$

$$x = \frac{2}{3}.$$

The tangent has an x-intercept at $x = \dfrac{2}{3}$.

Function, Tangent, and Normal



The tangent and normal meet at A and are perpendicular.

Finding the Tangent Line

Find the tangent to $y = \sin(x)$ at $x = \frac{\pi}{3}$.

$$y = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \qquad A = \left(\frac{\pi}{3}, \frac{\sqrt{3}}{2}\right),$$

$$\frac{dy}{dx} = \cos(x),$$

$$m_{\text{tangent}} = \left.\frac{dy}{dx}\right|_{x=\pi/3} = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}.$$

$$y - \frac{\sqrt{3}}{2} = \frac{1}{2}\left(x - \frac{\pi}{3}\right)$$

$$\approx \frac{x}{2} + 0.34$$

Finding the Normal Line

Find the normal to $y = \sin(x)$ at $x = \frac{\pi}{3}$.
The normal is perpendicular to the tangent, so

$$m_{\text{normal}} = -\frac{1}{m_{\text{tangent}}}$$

$$= -2.$$

Using $A = \left(\frac{\pi}{3}, \frac{\sqrt{3}}{2}\right)$,

$$y - \frac{\sqrt{3}}{2} = -2\left(x - \frac{\pi}{3}\right)$$

$$\approx -2x + 2.96$$

x-intercept of the Tangent Line

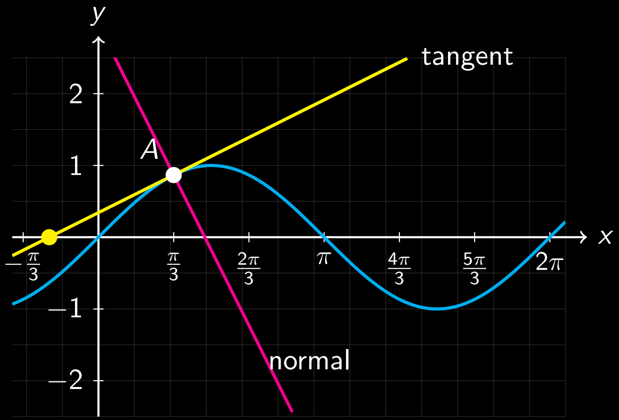
At the x-intercept of the tangent, $y = 0$:

$$\frac{x}{2} + 0.34 = 0$$

$$x = -0.68.$$

The tangent has an x-intercept at $x = -0.68$.

Function, Tangent, and Normal



$y = \sin(x)$, its tangent, and its normal at $x = \frac{\pi}{3}$.